

NUMPY

```
In [1]: my_list = [12,34,56]
        print(my_list)
```

[12, 34, 56]

```
In [2]: import numpy as np
        arr = np.array([12,34,56,78])
        print(arr)
```

[12 34 56 78]

USING PYTHON LIST

```
In [7]: my_list = [1,2,3]
        my_list2 = [4,5,6]
        total = []
        for i in range(len(my_list)):
            total.append(my_list[i] + my_list2[i])
        print(total)
```

[5, 7, 9]

USING NUMPY

```
In [11]: import numpy as np
         a = np.array([1,2,3])
         b = np.array([4,5,6])
         print(a+b)
         print(a-b)
         print(a*b)
         print(a/b)
         print(a**b)
```

```
[5 7 9]
[-3 -3 -3]
[ 4 10 18]
[0.25 0.4 0.5 ]
[ 1 32 729]
```

TYPES OF ARRAY AND ATTRIBUTES

```
In [24]: import numpy as np
#1D array
arr=np.array([1,2,3,4,5])
#2D array
arr1 = np.array([[1,2,3],[4,5,6]])
#multidimensional array
arr2 = np.array([[1,2,3],
                 [4,5,6],
                 [7,8,9]])

print(arr)
print(arr1)
print(arr2)
print(arr2.shape)
print(arr1.shape)
print(arr2.ndim)
print(arr1.ndim)
print(arr2.size)
print(arr1.size)
print(arr2.dtype)
```

```
[1 2 3 4 5]
[[1 2 3]
 [4 5 6]]
[[[1 2 3]
   [4 5 6]
   [7 8 9]]]
(1, 3, 3)
(2, 3)
3
2
9
6
int64
```

ARRAY CREATION IN NUMPY

```
In [60]: import numpy as np
a = np.array([1,2,3,4.5])
b = np.zeros(5)
print(a)
print(b)
c = np.zeros((2,2))
print(c)
d = np.ones(3)
print(d)
e = np.ones((2,3))
print(e)

print(np.arange(1,10,2))
print(np.linspace(0,20,5))
print(np.random.rand(5))
print(np.random.randint(1,100,10))
print(np.random.randn(5))
```

```
[1.  2.  3.  4.5]
[0.  0.  0.  0.  0.]
[[0.  0.]
 [0.  0.]]
[1.  1.  1.]
[[1.  1.  1.]
 [1.  1.  1.]]
[1  3  5  7  9]
[ 0.   5.  10.  15.  20.]
[0.47324491 0.10815809 0.07861896 0.35193487 0.65283788]
[93  7 29 57 49 81 72 93 16 77]
[ 0.2562272  0.17485032 -0.74178587 -1.16157225  0.34494809]
```

INDEXING AND SLICING

```
In [86]: #1d array indexing
import numpy as np
arr = np.array([10,20,35,48,50,60,70,80])
print(arr[0])
```

```
print(arr[-1])

#2d array indexing
arr2 = np.array([[10,20,30],
                 [40,50,60],
                 [22,33,44]])
print(arr2[1][1])

# 1D array slicing
print(arr[2:7])
print(arr[-7:-1])
print(arr[:5])
print(arr[2:])
print(arr[:,2])

# 2D array slicing
print(arr2[2])
print(arr2[:,2])

#boolean indexing
print(arr[arr>60])
print(arr[arr%2==0])

#fancy indexing
print(arr[[0,3,6]])

arr[2]=100
print(arr)
```

```

10
80
50
[35 48 50 60 70]
[20 35 48 50 60 70]
[10 20 35 48 50]
[35 48 50 60 70 80]
[10 35 50 70]
[22 33 44]
[30 60 44]
[70 80]
[10 20 48 50 60 70 80]
[10 48 70]
[ 10  20 100  48  50  60  70  80]

```

MATHEMATICAL OPERATIONS IN NUMPY

```

In [92]: #1D Scalar Broadcasting
import numpy as np
a = np.array([1,2,3])
print(a+10)

#2D Scalar Broadcasting
b = np.array([[1,2,3],
              [4,5,6]])
print(b+10)

#Row Broadcasting
#always no of columns must be same
print(b+a)

```

```

[11 12 13]
[[11 12 13]
 [14 15 16]]
[[2 4 6]
 [5 7 9]]

```

matrix operations

```

In [101... import numpy as np
a = np.array([[1,2],

```

```

        [4,5]])
b = np.array([[3,4],
              [5,6]])
print(a*b)
print(np.dot(a,b))
print(a@b)
print(np.matmul(a,b))

#Array Transpose
print(a.T) #Row becomes Column

```

```

[[ 3  8]
 [20 30]]
[[13 16]
 [37 46]]
[[13 16]
 [37 46]]
[[13 16]
 [37 46]]
[[1 4]
 [2 5]]

```

UNIVERSAL MATHEMATICAL FUNCTION

In [105...

```

import numpy as np
a = np.array([4,9,16])
print(np.sqrt(a))

#Absolute value
b = np.array([-4,-9,-16])
print(np.abs(b))

#Log values
print(np.log(a))

#exponential
print(np.exp(a))

```

```

[2.  3.  4.]
[ 4  9 16]
[1.38629436 2.19722458 2.77258872]
[5.45981500e+01 8.10308393e+03 8.8611052e+06]

```

```
In [8]: #Round function
import numpy as np
arr = np.array([1.234,5,674])
print(np.round(arr,2))
print(np.ceil(arr))
print(np.floor(arr))
```

```
[ 1.23  5. 674. ]
[  2.  5. 674.]
[  1.  5. 674.]
```

STATISTICAL FUNCTION

```
In [28]: #statistical function
import numpy as np
a = np.array([10,20,30,40,50])
print(np.mean(a))
print(np.median(a))
print(np.std(a))
print(np.max(a))
print(np.min(a))
print(np.sum(a))
print(np.percentile(a,25))
print(np.var(a))
print(np.argmin(a))
print(np.argmax(a))

b = np.array([1,2,3,4,5])
c = np.array([2,4,6,8,10])
print(np.corrcoef(b,c))
```

```
30.0
30.0
14.142135623730951
50
10
150
20.0
200.0
0
4
[[1. 1.]
 [1. 1.]]
```

AGGREGATE FUNCTION

```
In [32]: import numpy as np
a = np.array([[10,20,30],
              [40,50,60]])
print(np.sum(a))
print(np.sum(a,axis=0)) #Column wise
print(np.sum(a,axis=1)) #Row wise

210
[50 70 90]
[ 60 150]
```

```
In [52]: import numpy as np
scores = np.array([[45,67,89],
                   [60,70,80],
                   [90,40,80]])

print(scores)

print(np.mean(scores))

print(np.sum(scores,axis=0))
print(np.sum(scores,axis=1))

min_score = np.min(scores)
max_score = np.max(scores)

print(np.mean(scores,axis=0))
```

```
normalized_scores = (scores - min_score) / (max_score - min_score)
print(normalized_scores)
```

```
[[45 67 89]
 [60 70 80]
 [90 40 80]]
69.0
[195 177 249]
[201 210 210]
[65. 59. 83.]
[[0.1  0.54 0.98]
 [0.4  0.6  0.8 ]
 [1.   0.   0.8 ]]
```

```
In [59]: import numpy as np
arr = np.array([10,20,30,40,50])
exp = arr[arr>25]
print(exp)
print(sum(exp))
```

```
[30 40 50]
120
```

```
In [ ]:
```